Development of Styrofoam Cutter NC Machine for Intricate Cutting Path

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***Abstract*—This paper deals with the design, manufacture and application of NC machine for cutting 2D intricate path of plane styrofoam. The cutter uses nickel wire heated by DC electric current. The X and Y axis is moved using open loop controlled stepper motor. The path that is cut is first was taken from picture, than changed the captured picture into line graph, and then programming the machine to follow the resulted line. The result shows that the machine can cut well in intricate path shape in the form of wayang figure without typing manually long NC code. The testing shows that the average distance error reaches of 0,245 mm on average in both x and y axis.**

Keywords*—*NC machine; styrofoam cutter; intricate path; wayang figure

I. INTRODUCTION

Styrofoam usually refers to expanded polystyrene foam, often found in everyday life materials such as disposable coffee cups, coolers, or cushioning material in packaging. This material is typically white in color and is made of expanded polystyrene beads [1]. It was invented by a Swedish inventor, Carl Georg Munters [2]. Styrofoam is composed of 98% air, making it lightweight and buoyant. Because of its insulating properties and buoyancy, it also used for life raft or life vessels. Due to its low temperature conduction, the material usually used as insulating material, such as used in building insulation sheathing and pipe insulation. This materials also used to keep food remain cool, for example on the fish market [3]. For construction materials, styrofoam also used for light and strong wood composite [4].

Styrofoam is also often found in the community in the form of decorative display, information board, product billboard, substitute for balsa wood to make airplane aerofoil in aeromodelling, as well as for other uses.

In order to fulfill its function, this material must be formed in such a way as to obtain the expected shape. The method is to cut the material according to the desired shape. This cutting can be done in the form of two-dimensional cutting motion or 3D motion. This cutting process can be done manually by hand using a knife or nickelin electric wire. This method, of course, will not be effective if the cutting is in a considerable amount, or the styrofoam to be cut is thick, or if the desired shape is relatively complex and difficult to work by hand in a short time. The application of NC machine makes this cutting task can be done precisely and in higher speed.

Mechatronics technology on CNC machines is a technological application that works with high degree of automation. This machine is able to repeat many numbers of identical movements with high accuracy. The machine operates based on the NC code that is programmed by the operator on the control device. The disadvantage of this method is that the operators have to type dozens or even hundreds rows of NC codes. The rows may increase considerably when an intricate path should be followed.

To overcome these difficulties, it is necessary to create a CNC machine that can be operated without writing the complicated and tiring NC code lines. To do so, some works are using open source electronics and softwares such as [5], [6], and [7].

This paper presents the general idea to design, build and test electro-mechanical systems and software of a CNC Cartesian machine as an examples of implementation of open source system that can be used to cut styrofoam sheets in intricate paths.

II. MATERIALS AND METHOD

A. Forces on Nickel wire

The cutting of styrofoam, uses nickel wire heated by electric current. To cut styrofoam, it is needed that the temperature at the cutting surface is higher than 240^{o}C [8]. The forces on Nickel wire was measured to get to most appropriate cutting speed.

B. Guide Rail

Guide rail is made of cylindrical steel pipe. Teflon bearing was used to get a better glide on the rail. The deflection of the rail is kept small in order to get an accurate cutting. There are two guide rail for one axis. This is chosen to get a more reliable movement of the cutting tool.

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C. Coordinate system

The important thing to consider in designing the cartesian-driven equipment is the coordinate system. Determination of coordinate system is important especially in making the necessary software program. In cartesian coordinate systems it is commonly agreed that horizontal motion , this material is represented by the x-axis (horizontal) and vertical motion represented as the Y-axis (vertical). The common sign agreement is that the right and upward movements are positive, while in the opposite direction is negative. In practice if it is desired that the cutting tool only works in the first quadrant, then the initial position of the cutting tool is in the lower left corner i.e. at position 0.0. This is what will be applied to the design of this equipment. In the planning of this tool, the motion following the x-axis and the y-axis is done by stepper motors with simultaneous movement.

D. Movement of the Cutting Tool

To be able to pass the desired cutting motion, the cutting tool must be controlled to move along the trajectory of cutting motion according to the shape of the design of the object that was drawn using CAD software. This cutting path can be either straight or curved. In machines with computer-based controls such as plotter machines, printer machines, CNC machines and so on, there are known some standard codes that represent the movement of end effector to produce certain movement path. In this case the computer will translate the codes to control the machine. Some other codes are HPGL codes used on EPSON production plotter machines, gerber codes used in CAD / Computer Aided Design electronics, and G-Code codes used by mechanical engineering societies for Computerized Numerical Control machines.

In order to perform two-dimensional cutting motions the end effector is driven by two motion axes that represent the X-axis and Y-axis in the Cartesian coordinate system. In order for the cutting tool to perform a cutting motion with a path that is not parallel to the coordinate axis, for example a curve line, then the two axes of this motion must be controlled and driven simultaneously. In the discussion of machine tools technology motion control system like this is called the Contouring Control System.

In the Control system mentioned above, the tool is controlled by first determining the starting point and end point of the track line which must be passed by the cutting tool. This kind of thing is done by interpolation process. This control system is known as straight motion and circular curved motion. Straight motion is done by linear interpolation while circular motion is done by circular interpolation.

E. Development of NC code

The object to be cut is in the form of wayang figure named as *Werkudoro*. The path was created by first changed the picture of wayang figure into a collection of line graphs using image processing. These line graphs, then was somewhat repaired to get continuous paths. The resulted path then was changed to NC code with a free available software. The generated NC code should be adjusted in order to get a continuous path so that the wire can move continuously.

F. The Driving Motor

To move the X and Y axis, step motors were used as the driving motor. This kind of motor can be controlled in open loop manner without the need of position sensor.

III. RESULTS AND DISCUSSIONS

A. Forces on the cutting wire and cutting speed

Based on the tests performed, the forces that work in the cutting process vary depending on the speed, the slower the cutting speed the smaller the cutting force. At a certain velocity the cutting force is nearly zero. This is possible because at low velocity, the cut is not due to wire pressure on Styrofoam, but because of the melting of the styrofoam impregnated by the cutting wire. In this case the styrofoam passed by the wire will melt as it gets enough heat, thus forming a styrofoam cutting line.

Since the melting temperature of Styrofoam is around 240^{o}C, then the wire movement should wait until the Styrofoam reach the needed temperature. This waiting time can be expressed as cutting speed, where higher speed give a shorter waiting time. Figure 1 shows the varying wire force addition with the change in travel speed of the cutting wire, The result in this figure using wire diameter of 0,25 mm, voltage of 6 volt, current of 0,91 A and the wire resistant of 6,6 ohm at the wire length of 80 mm, and initial force of 10 N. The result shows that the appropriate cutting speed is around 1.7 cm/sec. The wire arrangement for the test is shown in Figure 2.

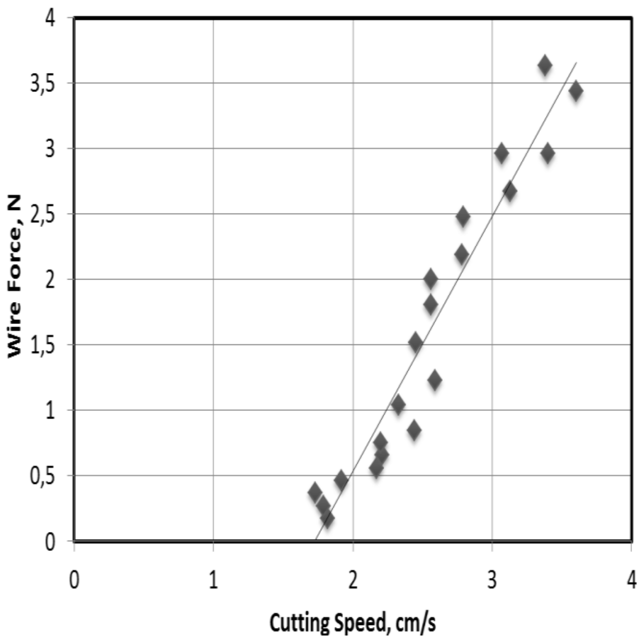

Figure 1. Wire force addition at various cutting speed

It can be seen that the speed can only be done on the certain value, since increasing the speed it will make the larger force in wire as well as on Styrofoam. This is because melting the

Styrofoam at its melting point needs certain time. To increase the cutting speed the temperature also should be increased in order to reach the melting temperature at shorter time. The temperature can be increased by increasing electric current. In the currently designed machine the speed of 1.7 cm/sec is the most appropriate.

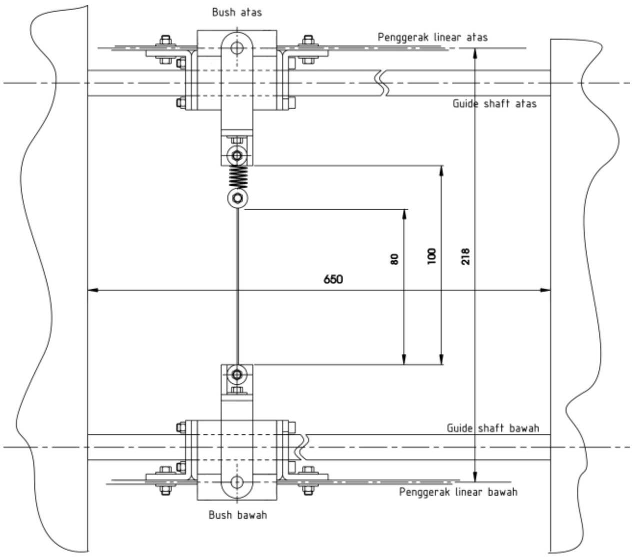

Figure 2. The wire arrangement on Y axis

B. *Forces on bearing bushing*

The cutter is moved in X and Y axis by a pair of stepper motor through toothed timing belt. The arrangement of X axis is shown in Figure 3. There are friction forces and cutting force that should be overcome by driving motors. Then the equation for motor force is shown in eq. 1.

$$F = K(\mu N + F_{cut}) \qquad (1)$$

Where K is belt system coefficient, μ is coeficient of friction in the hub, N is wire force, and F_{cut} is the cutting force.

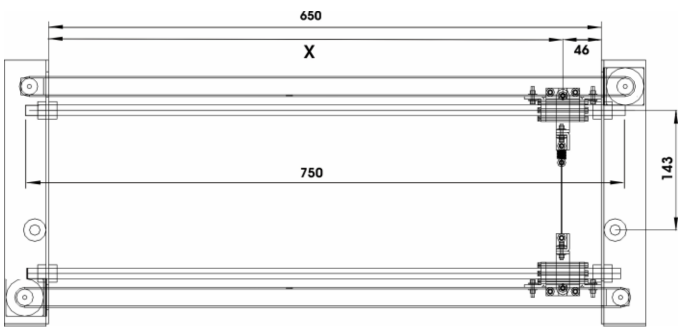

Figure 3. Driving arrangement on X axis.

C. *The programming design*

To get a CNC program, programmer usually writing the NC command, consists of G-code and M-code. In the intricate path, the path should be separated into many short part where each path should have its own NC code. Therefore, the programmer should type long NC code. In this research, the path that cut is in the form of wayang character named "*werkudoro"*, as shown in Figure 4. It can be seen that the path the cutter should follow is very intricate, consists of many contour and curve.

The NC program is performed as follows. First, the character was photographed to move the character to computer. Using an image processing program, the graph then was processed to get a line contour. In this step, however, the resulted line is often needs a manual repair. After a few repairing, it is then processed further to get the NC code. To do this, an open source freeware AceConverter was used. The process is shown in Figure 5.

The NC code that was used are limited to : G00, G01, G02, G03, G04, G05, G10, G11, G20, G21, G30, G31, G70, G71, G90, G91, M00, M01, M02, and M30.

Figure 4. A wayang character : Werkudoro

The translation from curve to NC code often gives some inaccurate and inappropriate result. Therefore it should be repaired for getting an better result.

D. *The motion accuracy*

The test of motion accuracy was done by repeating the 100 mm straight cutting movements a hundred times, then observed the distance error at each of cut line. The result is depicted in Figure 6. The average error is around 0,259 mm.

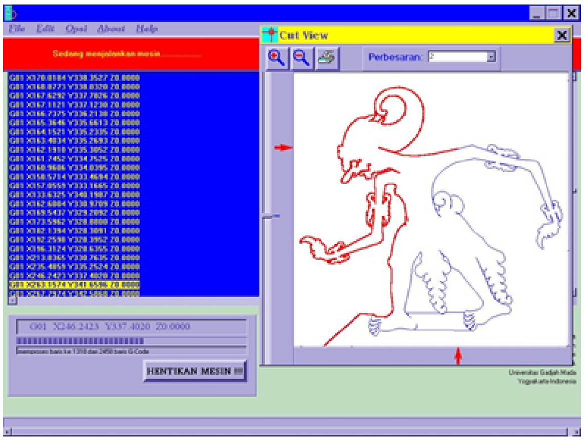

Figure 5. Changing the contour of wayang character into NC code

With the same method, the error of cutting distance on various directions is presented in Table 1. The test image of cutting hundreds mm straight lines is depicted in Figure 7. In the figure, the white part is styrofoam, and the black part is the blank.

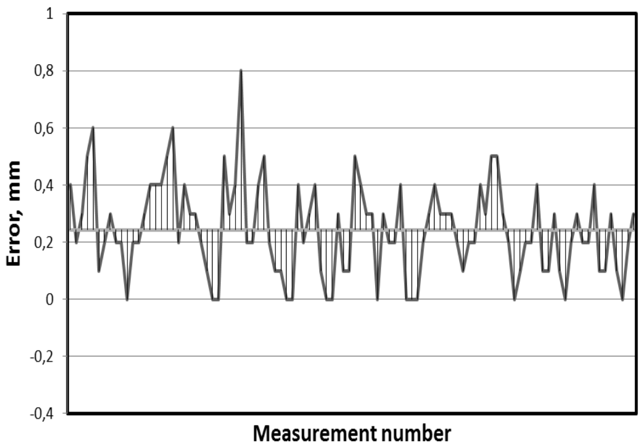

Figure 6. Error measurement on the 100 times cutting of 100 mm path.

Table 1. The Average distance error on various cutting direction.

Axis	Direction	Average Distance error
X	From right to left	0,263 mm
X	From left to right	0,231 mm
Y	From bottom to top	0,259 mm
Y	From top to bottom	0,227 mm

It can be seen that the error on all direction of cutting is less than 0,263 mm, or on average is 0,245 mm. This amount of error is usually small enough for general use of cut styrofoam.

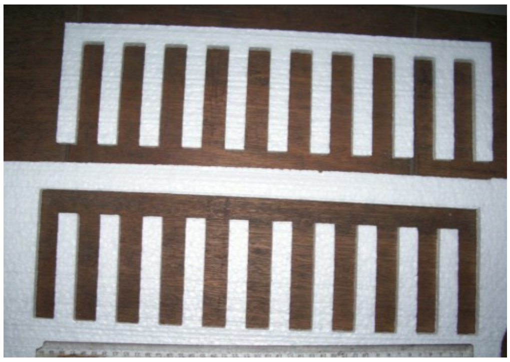

Figure 7. Cuts for accuracy test, hundreds mm cut.

IV. CONCLUSION

The use of hot wire cutter for cutting styrofoam plates in intricate path can be performed well through changing the picture into line graph and get the help of open source freeware of AceConverter to generate the complicated and intricate NC code. The result shows that the machine can cut well in intricate path shape in the form of wayang figure without typing manually long NC code. The testing shows that the machine has highest error at avalue of 0,263 mm, and on average of 0,245 mm.

REFERENCES

[1] "STYROFOAM™ - It's Not a Cup" Dow Chemical Company

[2] Boundy, Ray H.; Amos, J. Lawrence (1991). "A History of the Dow Chemical Physics Lab.", New York: Marcel Dekker, Inc. pp. 117–128. ISBN 0-8247-8097-3

[3] Rainer Froese, "Insulating properties of styrofoam boxes used for transporting live fish", Aquaculture, Volume 159, Issues 3–4, 1 January 1998, Pages 283-292

[4] Cenk Demirki, Semra Colak, Ismail Aydin, "Some technological properties of wood–styrofoam composite panels", Composites Part B: Engineering, Volume 55, December 2013, Pages 513–517

[5] Jorge E. Correa, , Nicholas Toombs, Placid M. Ferreira, A modular-architecture controller for CNC systems based on open-source electronics, Journal of Manufacturing Systems, Volume 44, Part 1

[6] Tomislav Staroveški, Danko Brezak, Toma Udiljak, Linuxcnc – The Enhanced Machine Controller: Application And An Overview, Tehnički vjesnik 20, 6(2013), 1103-1110

[7] L.M.Galantuccia, E.Piperib., F.Lavecchiaa., A.Zhavob, Semi-automatic Low Cost 3D Laser Scanning Systems for Reverse Engineering, Procedia CIRP, Volume 28, 2015, Pages 94-99

[8] http://styrofoam-densifier.com/physical-and-chemical-of-styrofoam-properties, downloaded 9 May 2017.